

Retail Store Sales & Customer Behaviour Analysis Analytics Report

Teyzix Core - Data Analytics Internship - Task DA (2)
End-to-End Data Analytics Project

1. Project Overview

This report documents an end-to-end retail analytics project carried out in the role of a Data Analyst for a retail company. The objective was to take raw, newly-collected business data and convert it into trustworthy insight: creating a realistic dataset, cleaning and transforming it, exploring it, and presenting the findings through an interactive dashboard and this report.

The project follows a complete analytics workflow — dataset creation, data cleaning, exploratory data analysis (EDA), insight generation, and dashboard development. The aim throughout was not simply to produce charts, but to identify the patterns and opportunities that would support real business decisions.

Tools Used

- Python (Pandas, NumPy) for data cleaning and exploratory analysis
- Matplotlib / Seaborn for visualisation
- Microsoft Power BI for the interactive dashboard

2. Dataset Description

A realistic retail sales dataset was created for this project, containing 1,700 transactions placed by 580 customers across six product categories and ten cities. The data spans 2024–2025. All monetary values are in Pakistani Rupees (Rs).

Field	Description
Transaction ID	Unique identifier for each sale
Customer ID	Identifier linking repeat purchases to a customer
Product Category	One of six categories (Electronics, Groceries, etc.)
Product Name	Specific product purchased
Quantity	Units bought in the transaction
Unit Price	Price per unit (Rs)
Total Amount	Quantity × Unit Price (Rs)
Purchase Date	Date of the transaction
Customer City	Customer location
Payment Method	Payment type used

3. Data Cleaning Process

The raw dataset was created with deliberate, realistic data-quality issues to simulate genuine business data. Each issue was identified and resolved using Python, with the reasoning documented below.

Issues Identified and Resolved

Issue	Resolution
Duplicate records (30 rows)	Removed with drop_duplicates()
Missing values (city, payment, quantity, price)	Numeric fields filled with median; categorical fields filled with "Unknown"
Inconsistent category labels	Standardised casing and whitespace; mapped variants to clean names
Inconsistent payment / city labels	Mapped all variants (e.g. "CC" → "Credit Card") to standard values
Mixed date formats	Parsed all formats into a single standard date type
Total Amount errors	Recalculated as Quantity × Unit Price
Impossible values (negative qty, zero price)	Treated as missing, then imputed

Outcome: the cleaned dataset contains 1,700 unique records with zero missing values, zero duplicates, six standardised categories, five payment methods, ten standardised cities, a single date format, and a Total Amount consistent with Quantity × Unit Price across every row.

4. Analysis Methodology

After cleaning, exploratory data analysis was performed to examine the business from several angles. Revenue, order counts, and average order value were aggregated across products, categories, cities, customers, payment methods, and time. The analysis focused on eight areas: monthly sales trends, top-selling products, category performance, revenue by city, customer purchasing patterns, payment-method distribution, average order value, and peak sales periods. Each was both quantified and visualised, and the results interpreted for their business meaning.

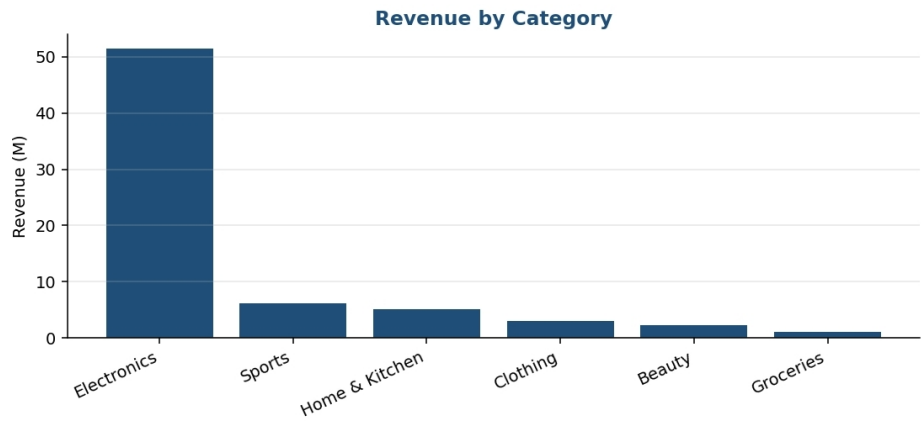
5. Visualizations

5.1 Monthly Sales Trend



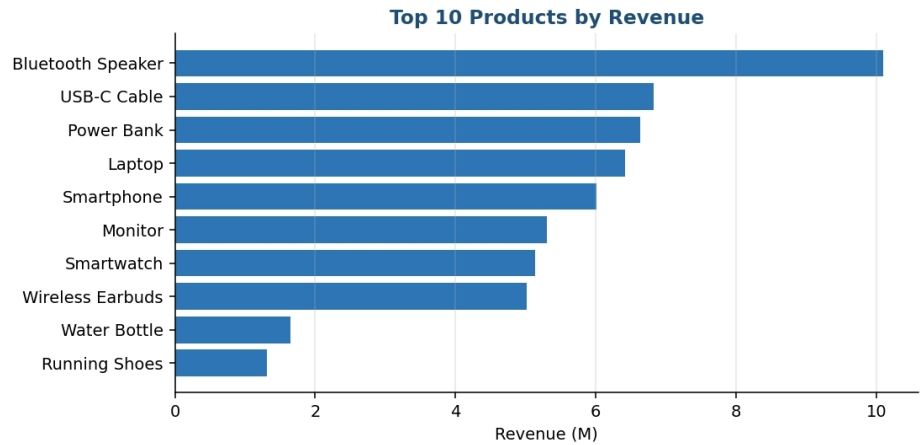
Revenue rises through the year, peaking in December 2024.

5.2 Revenue by Category



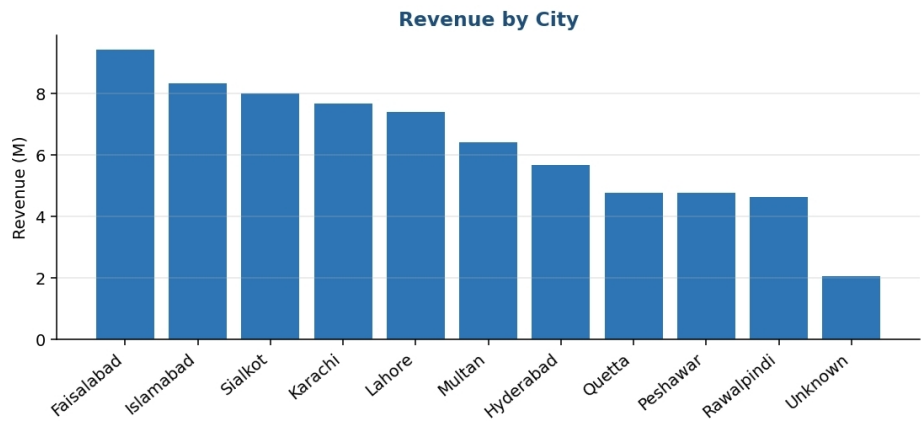
Electronics dominates total revenue.

5.3 Top 10 Products



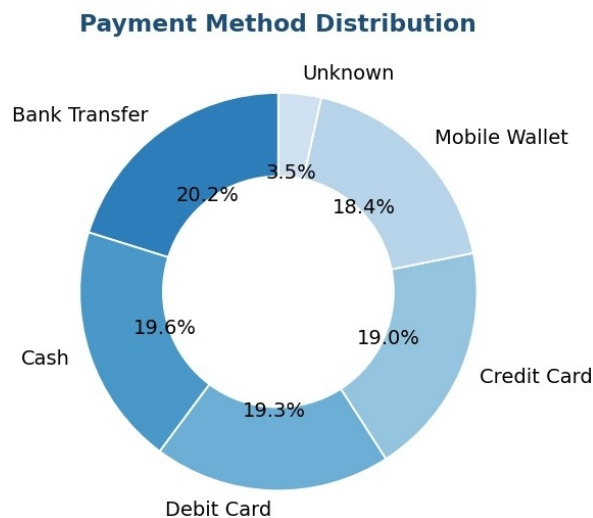
The Bluetooth Speaker leads; the top products are all Electronics.

5.4 Revenue by City



Revenue is spread fairly evenly; Faisalabad leads. "Unknown" denotes imputed missing data.

5.5 Payment Method Distribution



Payment methods are used in roughly equal proportion.

6. Key Insights

- **Revenue is driven by value, not volume.** Electronics produced 74% of revenue from under 17% of orders, with an average order of Rs178,523.
- **Groceries and Beauty under-contribute.** They sell frequently but at low value — high-traffic, low-revenue staples rather than failing lines.
- **Strong year-end seasonality.** Q4 revenue (Rs22.4M) is 75% higher than Q1, peaking in December; Saturday is the strongest day.
- **An exceptionally loyal customer base.** 79% of customers are repeat buyers, and repeat customers generate 92% of all revenue.
- **Concentration risk.** Heavy reliance on one category (Electronics) leaves the business exposed if that category weakens.

7. Recommendations

1. **Capitalise on Q4 seasonality** — concentrate inventory, staffing, and promotions in October–December and address the weak Q1.
2. **Protect Electronics while diversifying** — grow Sports and Home & Kitchen as secondary revenue pillars to reduce concentration risk.
3. **Launch a loyalty programme** — with 92% of revenue from repeat customers, retention is the highest-return investment available.
4. **Cross-sell on frequent trips** — attach higher-value items to frequent Grocery and Beauty purchases to lift the average basket.

5. **Reinforce the weekend peak** — align promotions and staffing with the strong Saturday sales pattern.